CNS MODULE 2 QB SOLVED

Question 1 – Explain the services provided by the Network Layer in the OSI Model.

Answer (8 Marks)

Introduction

The Network Layer is the third layer of the OSI (Open Systems Interconnection) model. It is responsible for delivering packets from the source host to the destination host across multiple interconnected networks.

The network layer provides the following services:

Services of the Network Layer

1. Packetizing

- Packetizing is the process of encapsulating the data received from the upper layer into a network-layer packet.
- At the source, the network layer adds a header containing the source address, destination address, and other required information.
- At the destination, the packet is decapsulated, and the original data is delivered to the upper layer.
- If packets are fragmented, the network layer is responsible for reassembling them at the destination.

2. Routing

- Routing is the process of finding the best path for a packet from the source to the destination.
- A network consists of multiple interconnected LANs, WANs, and routers.
- Since multiple paths may exist, the network layer uses routing algorithms to determine the best route.
- Examples of routing algorithms include Distance Vector and Link State routing.

3. Forwarding

- Forwarding is the process of transferring a packet from an incoming interface to the appropriate outgoing interface of a router.
- The router uses a forwarding table or routing table to make forwarding decisions.
- The destination network address is used to determine the outgoing interface.
- Forwarding is performed at every router along the path.

4. Error Control

- Error control is used to detect corruption in network-layer packets.
- The IPv4 datagram contains a checksum field to detect errors in the header.
- The checksum helps detect changes or corruption in the header during transmission.
- Error checking of the entire datagram is not performed by IPv4.

5. Flow Control

- Flow control regulates the amount of data sent by a source so that the receiver is not overwhelmed.
- It is required when the source produces data faster than the destination can process it.
- The receiver provides feedback to the sender when it cannot handle the incoming data.
- This helps prevent the receiver from being overloaded.

6. Congestion Control

- Congestion occurs when the number of datagrams sent into the network exceeds the capacity of the network or routers.
- During congestion, routers may drop packets.
- The sender may retransmit lost packets, which can further increase congestion.
- Congestion control helps prevent excessive traffic and improves network performance.

7. Quality of Service (QoS)

- Quality of Service refers to providing better communication quality for different types of applications.
- It is particularly important for multimedia applications such as audio and video communication.
- QoS helps support applications that require better network performance.

- It is important for reliable and efficient multimedia communication.

8. Security

- The network layer was originally designed without security provisions.
- To provide security for the connectionless network layer, IP Security (IPSec) was introduced.
- IPSec provides security services for IP communication.
- It can provide authentication, data integrity, and confidentiality.

Question 2 – What is Packet Switching? Explain the Virtual-Circuit Approach to Packet Switching.

Answer (8 Marks)

1. Packet Switching

Packet switching is a method of transmitting data in which a message from the upper layer is divided into smaller units called packets.

- Each packet is transmitted individually through the network.
- Routers are used to forward packets from the source to the destination.
- The destination receives the packets and waits until all packets belonging to the same message arrive.
- The packets are then reassembled, and the original message is delivered to the upper layer.

At the network layer, packet switching is used because the unit of data is a packet.

2. Virtual-Circuit Approach

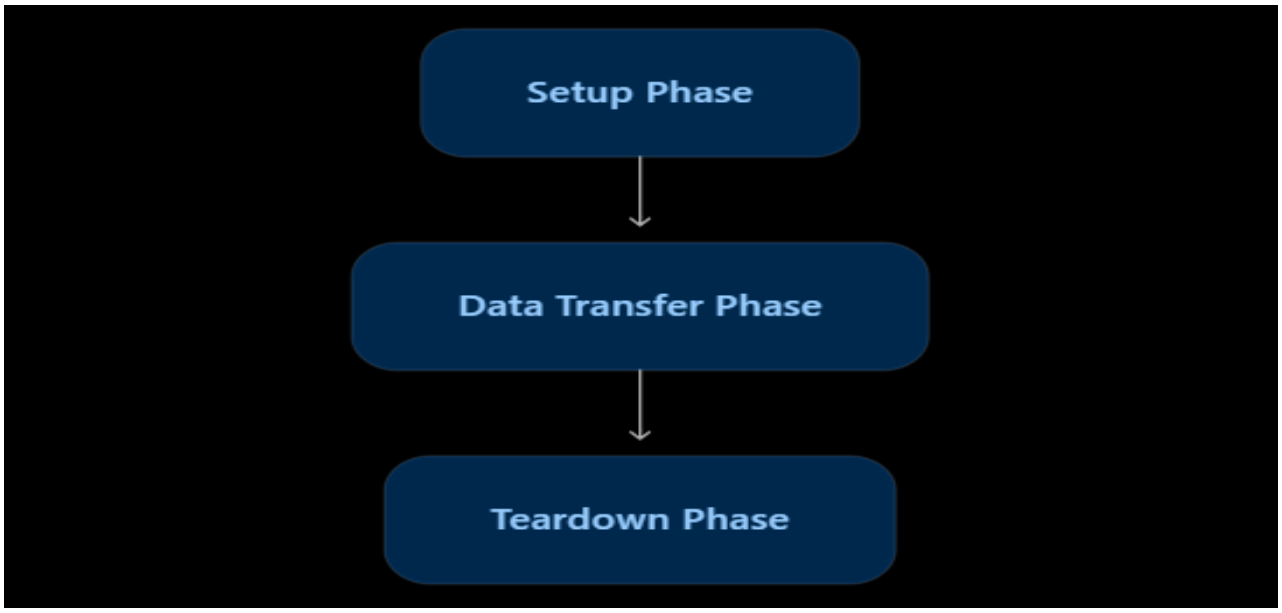
The virtual-circuit approach is a connection-oriented service in which all packets belonging to a message follow a predetermined path.

Before sending packets, a virtual connection must be established between the source and destination.

- All packets belonging to the same message follow the same path.
- Each packet is forwarded based on a label present in its header.
- Routers maintain table entries for the virtual circuit.
- The communication takes place in three phases.

3. Three Phases of Virtual-Circuit Approach

The virtual-circuit approach consists of three phases:



A. Setup Phase

- Before data transmission, a virtual circuit is established between the source and destination.
- The source sends a request packet containing the source and destination addresses.
- Each router along the path creates an entry in its switching table.
- The destination assigns a label to the incoming packets.
- An acknowledgment packet is sent back to complete the table entries.

B. Data Transfer Phase

- After the virtual circuit is established, the source begins transmitting data packets.
- All packets follow the same predetermined path.
- Each router uses the label in the packet to determine the outgoing port and label.

- The packets are delivered to the destination in sequence.

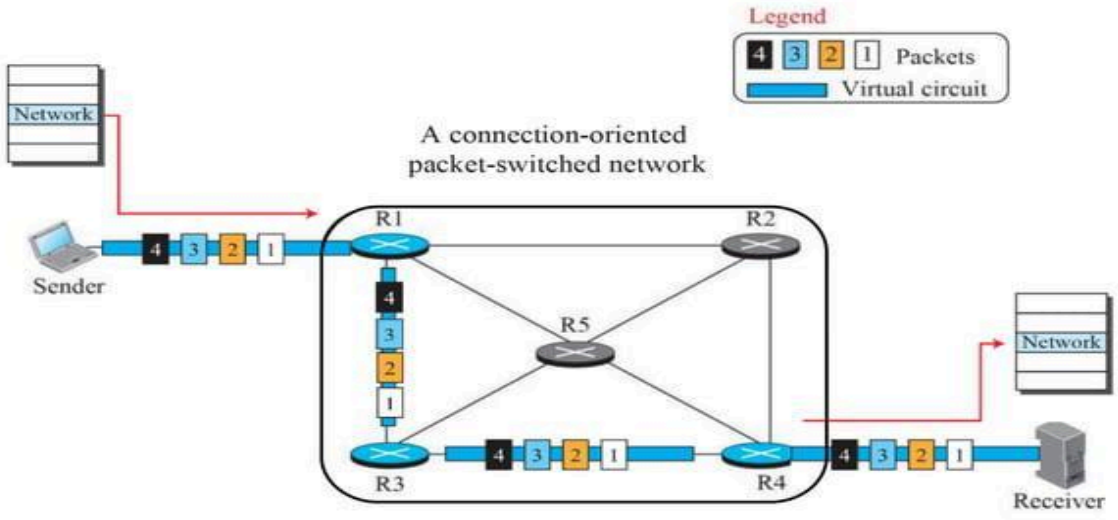
C. Teardown Phase

- After all packets have been transmitted, the source sends a special packet called a teardown packet.
- The destination responds with a confirmation packet.
- All routers delete the corresponding virtual-circuit entries from their switching tables.
- The virtual connection is terminated.

4. Example of Virtual-Circuit Packet Switching

Consider a virtual connection established between source A and destination B through routers R1, R3, and R4.

Figure 18.5: A virtual-circuit packet-switched network



18.13

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The packets travel through the same path:

A → R1 → R3 → R4 → B

Each router forwards the packet using the label stored in its switching table.

For example, a router may use the following table:

Incoming port	Incoming label	Outgoing port	Outgoing label
1	23	3	66

The router receives a packet with label 23 on port 1 and forwards it through port 3 with label 66.

Question 3. Explain the Performance Parameters of the Network Layer, Including Delay, Throughput, and Packet Loss

Introduction

The performance of a network layer is measured by how efficiently it delivers packets from the source to the destination. The main performance parameters are delay, throughput, and packet loss.

1. Delay

Delay is the time taken by a packet to travel from the source to the destination.

The total delay consists of four components:

- Processing delay: Time taken by a router to examine the packet header and make a forwarding decision.
- Queuing delay: Time a packet waits in the router's queue before transmission.
- Transmission delay: Time required to place all packet bits onto the link.
- Propagation delay: Time taken by the signal to travel through the physical medium.

Total delay is calculated as:

$$D = D_{processing} + D_{queuing} + D_{transmission} + D_{propagation}$$

Example: If processing delay = 2 ms, queuing delay = 5 ms, transmission delay = 3 ms, and propagation delay = 10 ms:

$$D = 2 + 5 + 3 + 10 = 20ms$$

A lower delay means faster packet delivery.

2. Throughput

Throughput is the amount of data successfully delivered from the source to the destination per unit time.

It is measured in bits per second (bps), Kbps, or Mbps.

$$Throughput = \frac{Datasuccessfullydelivered}{Totaltimetaken}$$

Example: If 10 megabits of data are successfully delivered in 2 seconds:

$$Throughput = \frac{10}{2} = 5Mbps$$

Higher throughput indicates better network performance. Throughput depends on bandwidth, congestion, packet loss, and network conditions.

3. Packet Loss

Packet loss occurs when one or more packets fail to reach their destination.

Packets may be lost due to:

- Network congestion.
- Router buffer overflow.
- Faulty network links.
- Transmission errors.

Packet loss is usually expressed as a percentage.

$$\text{Packet Loss (\%)} = \frac{\text{Packets lost}}{\text{Total packets sent}} \times 100$$

Example: If 100 packets are sent and 5 packets are lost:

$$\text{Packet Loss} = \frac{5}{100} \times 100 = 5\%$$

High packet loss can cause retransmissions, reduced throughput, and poor video or voice quality.

Summary

Parameter	Meaning	Desired condition
Delay	Time taken for packet delivery	Low
Throughput	Data delivered per unit time	High
Packet loss	Packets that fail to reach destination	Low

Question 4. Explain IPv4 Addressing and Describe the Different Classes of IPv4 Addresses

1. IPv4 Addressing

An IPv4 address is a 32-bit address that uniquely and universally identifies the connection of a host or router to the Internet.

- IPv4 address consists of 32 bits.
- The total address space is 2^{32}, which is 4,294,967,296 addresses.
- It is usually represented in dotted-decimal notation.

- The 32 bits are divided into four groups of 8 bits called octets.
- Each octet is represented by a decimal number from 0 to 255.

Example:

IP address: `192.168.1.10`

Binary representation:

`11000000.10101000.00000001.00001010`

An IPv4 address is divided into two parts:

- Prefix (Network ID): Identifies the network.
- Suffix (Host ID): Identifies a particular host or connection within the network.

2. Classes of IPv4 Addresses

In classful addressing, the IPv4 address space is divided into five classes: A, B, C, D, and E. Classes A, B, and C are used for different network sizes, Class D is used for multicast, and Class E is reserved.

Class	Starting bits	First-octet range	Network/Host division	Purpose
A	`0`	1–126*	8-bit network, 24-bit host	Large networks
B	`10`	128–191	16-bit network, 16-bit host	Medium-sized networks
C	`110`	192–223	24-bit network, 8-bit host	Small networks
D	`1110`	224–239	Not divided into network and host	Multicast

E	`1111`	240–255	Not divided into network and host	Reserved

*The range 127 is reserved for loopback purposes, and 0 is reserved for special use.

Class A

- Designed for large networks.
- The first bit is always `0`.
- The first 8 bits represent the network ID.
- The remaining 24 bits represent the host ID.
- The first-octet range is 1 to 126.

Example: `10.1.2.3`

Class B

- Designed for medium-sized networks.
- The first two bits are `10`.
- The first 16 bits represent the network ID.
- The remaining 16 bits represent the host ID.
- The first-octet range is 128 to 191.

Example: `172.16.5.10`

Class C

- Designed for small networks.
- The first three bits are `110`.
- The first 24 bits represent the network ID.
- The remaining 8 bits represent the host ID.
- The first-octet range is 192 to 223.

Example: `192.168.1.10`

Class D

- The first four bits are `1110`.
- The address range is 224.0.0.0 to 239.255.255.255.

- It is used for multicast addressing.
- It is not divided into network ID and host ID.

Class E

- The first four bits are 1111.
- The address range is 240.0.0.0 to 255.255.255.255.
- It is reserved for experimental and future use.
- It is not divided into network ID and host ID.

3. Important Points

- IPv4 uses a 32-bit address.
- Classful addressing uses fixed-length network prefixes: 8, 16, or 24 bits for Classes A, B, and C.
- Class D is used for multicast.
- Class E is reserved.
- Classful addressing is now largely obsolete and has been replaced by classless addressing (CIDR), which uses variable-length prefixes.

Question 5. Explain Class A, B, C, D, and E IPv4 Address Classes. What Are the Disadvantages of Classful Addressing?

1. Classful Addressing

In classful addressing, the 32-bit IPv4 address space is divided into five classes: A, B, C, D, and E.

The class of an IPv4 address is determined by its leading bits. Classes A, B, and C are divided into network and host portions, whereas Classes D and E are not divided into these portions.

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2. Different Classes of IPv4 Addresses

Class	Leading bits	First-octet range	Network bits	Host bits	Purpose
A	`0`	1–126*	8	24	Large networks
B	`10`	128–191	16	16	Medium-sized networks
C	`110`	192–223	24	8	Small networks
D	`1110`	224–239	—	—	Multicast
E	`1111`	240–255	—	—	Reserved/experimental

The first-octet value 0 is reserved, and 127 is used for loopback.

Class A

- The first bit is always `0`.
- The first 8 bits represent the network ID.
- The remaining 24 bits represent the host ID.
- It is designed for large networks.
- Example: `10.1.2.3`

Class B

- The first two bits are `10`.
- The first 16 bits represent the network ID.
- The remaining 16 bits represent the host ID.
- It is designed for medium-sized networks.
- Example: `172.16.5.10`

Class C

- The first three bits are `110`.
- The first 24 bits represent the network ID.
- The remaining 8 bits represent the host ID.
- It is designed for small networks.
- Example: `192.168.1.10`

Class D

- The first four bits are `1110`.
- The address range is 224.0.0.0 to 239.255.255.255.
- It is used for multicast addressing.
- It identifies a group of hosts rather than an individual host.

Class E

- The first four bits are `1111`.
- The address range is 240.0.0.0 to 255.255.255.255.
- It is reserved for experimental and research purposes.
- It is not used for normal host addressing.

3. Disadvantages of Classful Addressing

Classful addressing has the following disadvantages:

1. Wastage of IP addresses:
The fixed sizes of Class A, B, and C networks may allocate more addresses than an organization requires. For example, a Class B network provides a large host address space even if the organization needs only a few thousand addresses.

2. Inefficient address allocation:
Organizations cannot request a network block of an arbitrary size. They must choose from fixed class sizes.

3. Address depletion:
Due to inefficient allocation and wastage, IPv4 addresses are consumed rapidly, leading to address depletion.

4. Lack of flexibility:
The network and host portions are fixed at 8, 16, or 24 bits for Classes A, B, and C. The network prefix cannot be adjusted freely according to requirements.

5. Large routing tables:
Classful addressing does not support flexible aggregation as effectively as classless addressing, which can increase the size of routing tables.

Question 6. Explain CIDR (Classless Inter-Domain Routing) with an Example. How Does CIDR Reduce IP Address Wastage?

1. Definition of CIDR

CIDR (Classless Inter-Domain Routing) is an IP addressing method that uses variable-length network prefixes instead of fixed network prefixes used in classful addressing.

- CIDR removes the fixed boundaries of Classes A, B, and C.
- It allows IP addresses to be allocated in variable-sized blocks according to the requirements of an organization.
- It uses slash notation to represent the prefix length.
- The prefix identifies the network, and the remaining bits identify the host.

2. CIDR Notation

In CIDR, the prefix length is written after the IP address using a slash (`/`).

Example: `192.168.1.0/24`

Here:

- `192.168.1.0` is the IP address.
- `/24` indicates that the first 24 bits represent the network prefix.
- The remaining 8 bits represent the host portion.

The number of addresses in a CIDR block is calculated using:

$$N = 2^{32-n}$$

where:

- N = number of addresses in the block
- n = prefix length

3. Example of CIDR

Consider the address block:

`192.168.1.0/26`

The prefix length is 26.

Number of addresses:

$$N = 2^{32-26}$$

$$N = 2^{6} = 64$$

Therefore, the block contains 64 addresses.

The address range is:

- First address: `192.168.1.0`
- Last address: `192.168.1.63`

The 26 leftmost bits identify the network, and the remaining 6 bits identify addresses within the block.

4. How CIDR Reduces IP Address Wastage

In classful addressing, an organization must select a fixed-size block such as Class C, which contains 256 addresses.

However, an organization may require only 64 addresses.

In classless addressing, the organization can be allocated a block of 64 addresses using `/26`.

Addressing method	Addresses allocated	Addresses required	Wastage
Classful (Class C)	256	64	192

CIDR (/26)	64	64	0

Thus, CIDR reduces IP address wastage by allocating address blocks according to the actual requirements of organizations instead of using fixed class sizes.

CIDR also allows address aggregation, in which multiple smaller address blocks are combined into a larger block. This helps reduce the number of entries in routing tables.

5. Advantages of CIDR

1. Reduces IP address wastage.
2. Supports variable-length network prefixes.
3. Allows flexible allocation of IP address blocks.
4. Helps overcome IPv4 address depletion.
5. Reduces routing table size through address aggregation.

Question 7. Differentiate Between Classful and Classless Addressing. Explain CIDR with an Example.

1. Classful Addressing

Classful addressing is an IP addressing method in which IP addresses are divided into fixed classes such as Class A, Class B, and Class C.

Each class has a fixed network and host portion.

2. Classless Addressing

Classless addressing is an IP addressing method in which IP addresses are allocated using variable-length network prefixes.

It does not follow the fixed boundaries of Class A, Class B, and Class C. CIDR is the technique used for classless addressing.

3. Difference Between Classful and Classless Addressing

Classful Addressing	Classless Addressing
IP addresses are divided into fixed classes.	IP addresses are allocated without fixed classes.
Uses Classes A, B, C, D, and E.	Does not use address classes for allocation.
Network prefix length is fixed for each class.	Network prefix length is variable.
Uses default subnet masks.	Uses CIDR notation with a prefix length.
Does not support flexible address allocation.	Supports flexible address allocation.
Can cause more IP address wastage.	Reduces IP address wastage.
Routing updates may contain class-based network information.	Routing updates include prefix length information.

4. CIDR (Classless Inter-Domain Routing)

CIDR is an IP addressing method that uses variable-length network prefixes instead of fixed network prefixes.

- It removes the fixed boundaries of Classes A, B, and C.
- It allows IP addresses to be allocated in variable-sized blocks.
- It uses slash notation to represent the network prefix.
- The prefix identifies the network, and the remaining bits identify the host.

5. CIDR Notation

In CIDR, the prefix length is written after the IP address using a slash (`/`).

Example: 192.168.1.0/24

Here:

- `192.168.1.0` is the IP address.
- `/24` indicates that the first 24 bits represent the network prefix.
- The remaining 8 bits represent the host portion.

The number of addresses in a CIDR block is calculated using:

$$N = 2^{32-n}$$

Where:

- N = number of addresses in the block
- n = prefix length

6. Example of CIDR

Consider the address block:

192.168.1.0/26

The prefix length is 26.

Number of addresses:

$$N = 2^{32-26}$$

$$N = 2^{6} = 64$$

Therefore, the block contains 64 addresses.

The address range is:

- First address: `192.168.1.0`
- Last address: `192.168.1.63`

The 26 leftmost bits identify the network, and the remaining 6 bits identify addresses within the block.

7. How CIDR Reduces IP Address Wastage

In classful addressing, an organization may be allocated a Class C block containing 256 addresses, even if it requires only 64 addresses.

Using CIDR, the organization can be allocated a `/26` block containing 64 addresses.

Addressing Method	Addresses Allocated	Addresses Required	Address Wastage
Classful (Class C)	256	64	192
CIDR (/26)	64	64	0

Thus, CIDR reduces IP address wastage by allocating address blocks according to the actual requirements of organizations.

CIDR also supports address aggregation, where multiple smaller address blocks are combined into a larger block. This helps reduce the number of entries in routing tables.

8. Advantages of CIDR

1. Reduces IP address wastage.
2. Supports variable-length network prefixes.
3. Allows flexible allocation of IP address blocks.
4. Helps overcome IPv4 address depletion.
5. Reduces routing table size through address aggregation.

Question 8. Explain the Format of an IPv4 Datagram and the Functions of Its Major Header Fields.

1. Definition of an IPv4 Datagram

An IPv4 datagram is a packet used by the Internet Protocol (IPv4) to transfer data from a source host to a destination host.

It consists of two main parts:

- Header: Contains control information required for delivering the packet.
- Data (Payload): Contains the actual data being transmitted.

The IPv4 header has a minimum size of 20 bytes and a maximum size of 60 bytes.

2. Format of an IPv4 Datagram

An IPv4 datagram consists of a header and data.

- Header size: 20 to 60 bytes
- Data: Variable length
- Maximum total datagram size: 65,535 bytes

The IPv4 header contains the following fields:

Field	Size	Function
Version	4 bits	Specifies the IP version. For IPv4, the value is 4.
Header Length (IHL)	4 bits	Specifies the length of the header. The minimum is 20 bytes and the maximum is 60 bytes.
Type of Service (ToS)	8 bits	Specifies the service requirements of the datagram, such as priority and quality of service.

Total Length	16 bits	Specifies the total length of the datagram, including the header and data.
Identification	16 bits	Identifies fragments belonging to the same original datagram.
Flags	3 bits	Controls fragmentation. It indicates whether fragmentation is allowed and whether more fragments follow.
Fragment Offset	13 bits	Specifies the position of a fragment in the original datagram.
Time to Live (TTL)	8 bits	Limits the lifetime of a datagram. Each router decreases the TTL value by at least 1. If it reaches 0, the datagram is discarded.
Protocol	8 bits	Identifies the upper-layer protocol, such as TCP or UDP, to which the data must be delivered.
Header Checksum	16 bits	Detects errors in the IPv4 header. It is recalculated at each router because some header fields may change.
Source Address	32 bits	Specifies the IPv4 address of the sender.
Destination Address	32 bits	Specifies the IPv4 address of the receiver.

Options	Variable	Provides additional features such as record route and timestamp.
Padding	Variable	Adds extra bits to make the header length a multiple of 32 bits.

3. Functions of Major Header Fields

1. Version

The 4-bit version field specifies the IP version used. For IPv4, the value is 4.

2. Header Length (IHL)

The IHL field specifies the length of the IPv4 header. The minimum header length is 20 bytes, and the maximum is 60 bytes.

3. Total Length

The total length field specifies the complete size of the datagram, including both the header and the data.

4. Identification, Flags, and Fragment Offset

These fields are used for fragmentation and reassembly.

- Identification: Identifies fragments belonging to the same datagram.
- Flags: Indicates fragmentation-related information.
- Fragment Offset: Specifies the position of a fragment in the original datagram.

5. Time to Live (TTL)

TTL prevents a datagram from circulating indefinitely in the network. Each router decreases the TTL value. When TTL becomes 0, the datagram is discarded.

6. Protocol

The protocol field identifies the upper-layer protocol.

Examples:

- TCP = 6
- UDP = 17

7. Header Checksum

The checksum is used to detect errors in the IPv4 header. It does not check the data portion of the datagram.

8. Source and Destination Addresses

- Source address identifies the sender.
- Destination address identifies the receiver.

These addresses help routers forward the datagram toward its destination.

Question 9. Explain Fragmentation and Reassembly in IPv4. Why Are They Required?

1. Definition of Fragmentation

Fragmentation is the process of dividing a large IPv4 datagram into smaller pieces called fragments so that they can travel through a network.

Each network has a Maximum Transmission Unit (MTU), which specifies the maximum size of a packet that can be transmitted through that network.

If the size of an IPv4 datagram is greater than the MTU of a network, fragmentation may be required.

2. Why Is Fragmentation Required?

Fragmentation is required because:

1. Different networks have different MTU sizes.
2. A datagram generated by the source may be larger than the MTU of an intermediate network.
3. Smaller fragments can be transmitted through networks that cannot carry the original large datagram.
4. It helps IPv4 datagrams travel across networks with different packet-size limitations.

3. Fragmentation in IPv4

When an IPv4 datagram is larger than the MTU of a network, the source or an intermediate router divides it into smaller fragments.

Each fragment contains:

- Its own IPv4 header.
- A portion of the original data.
- The same Identification value as the original datagram.
- A Fragment Offset indicating its position in the original datagram.
- A More Fragments (MF) flag indicating whether additional fragments follow.

Important fragmentation fields:

Field	Function
Identification	Identifies fragments belonging to the same original datagram.
Flags	Indicates whether more fragments follow.
Fragment Offset	Specifies the position of the fragment in the original data.

4. Example of Fragmentation

Consider an IPv4 datagram with:

- Total datagram size = 4000 bytes
- Header size = 20 bytes
- Data size = 3980 bytes
- MTU of the network = 1500 bytes

Each fragment must fit within 1500 bytes, including its 20-byte header.

Maximum data per fragment = 1500 − 20 = 1480 bytes.

The original datagram is divided into three fragments:

Fragment	Data Size	Total Size	Fragment Offset	MF Flag
1	1480 bytes	1500 bytes	0	1
2	1480 bytes	1500 bytes	185	1
3	1020 bytes	1040 bytes	370	0

Explanation:

- The first fragment carries the first 1480 bytes of data.
- The second fragment carries the next 1480 bytes.
- The third fragment carries the remaining 1020 bytes.
- The MF flag is 1 for the first two fragments because more fragments follow.
- The MF flag is 0 for the last fragment because no more fragments follow.

The Fragment Offset is measured in units of 8 bytes.

For example:

Fragment 2 offset = 1480 / 8 = 185

Fragment 3 offset = 2960 / 8 = 370

5. Reassembly in IPv4

Reassembly is the process of combining the received fragments to reconstruct the original IPv4 datagram.

Reassembly takes place at the destination host, not at intermediate routers.

The process is as follows:

1. The destination receives the fragments.
2. It identifies fragments belonging to the same datagram using the Identification field.

3. It uses the Fragment Offset field to determine the correct position of each fragment.
4. It arranges the fragments in the correct order.
5. It combines the fragments to reconstruct the original datagram.
6. The reconstructed datagram is delivered to the upper-layer protocol.

If one or more fragments are missing, the destination cannot completely reconstruct the original datagram. After the reassembly timer expires, the incomplete datagram is discarded.

6. Difference Between Fragmentation and Reassembly

Fragmentation	Reassembly
Divides a large datagram into smaller fragments.	Combines fragments into the original datagram.
Performed at the source or intermediate routers in IPv4.	Performed at the destination host.
Uses Identification, Flags, and Fragment Offset fields.	Uses Identification and Fragment Offset fields to arrange fragments.
Required when the datagram exceeds the network MTU.	Required to recover the original datagram.

Question 10. Explain the Distance-Vector Routing Algorithm with a Suitable Example

Definition

The Distance-Vector Routing (DVR) algorithm is a dynamic routing algorithm in which each router maintains a routing table containing the best-known distance (cost) to every destination and the next router to reach it.

Each router shares its routing table only with its directly connected neighbours. The algorithm is based on the Bellman-Ford algorithm.

Working of the Algorithm

1. Initialization: Each router sets its distance to itself as 0, the cost to directly connected neighbours as the link cost, and the distance to all other routers as infinity (∞).
2. Exchange of information: Each router periodically sends its distance vector to its immediate neighbours.
3. Updating the routing table: When a router receives a neighbour's distance vector, it calculates the cost of reaching each destination through that neighbour.
4. Choosing the shortest path: The router selects the route with the minimum total cost and updates its routing table.
5. Convergence: This process continues until no router needs to change its routing table.

Bellman-Ford Equation

The distance to destination `y` from router `x` is calculated as:

$$D_x(y) = \min_v \{c(x,v) + D_v(y)\}$$

Where:

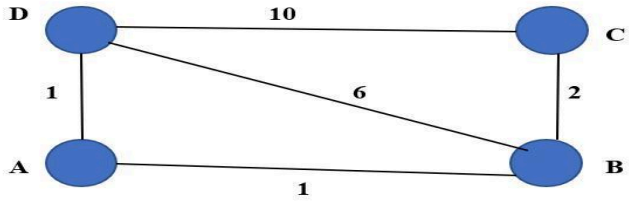
- $D_x(y)$ = minimum cost from router `x` to destination `y`
- $c(x,v)$ = cost of the link from router `x` to its neighbour `v`
- $D_v(y)$ = cost from neighbour `v` to destination `y`
- `v` = a directly connected neighbour of router `x`

In simple words, the cost through a neighbour = cost to reach that neighbour + the neighbour's cost to reach the destination. The router chooses the smallest total cost.

Suitable Example

Consider the following network with four routers and the given link costs:

Consider the following network running the **distance vector routing protocol**. In the diagram, vertices represent routers and edges (arcs) represent links between routers. The numerical entries on the links represents link costs. Higher costs indicate worse links.
Show the routing tables at nodes A, B, C, and D, when the distance vector routing algorithm stabilizes. Routing tables should contain To (Destination), Cost and Next Hop as fields.
(2 marks)



The link costs are:

- A–B = 1
- B–C = 2
- C–D = 1
- A–D = 8

Step 1: Initial information at router A

Router A initially knows only its direct connections:

- Distance from A to A = 0
- Distance from A to B = 1
- Distance from A to D = 8
- Distance from A to C = ∞

Step 2: Router A receives information from neighbour B

Router B has learned that it can reach router D through C with a cost of:

$$B \rightarrow C \rightarrow D = 2 + 1 = 3$$

Router A calculates the cost of reaching D through B:

$$A \rightarrow B \rightarrow C \rightarrow D$$

$$= 1 + 3 = 4$$

A also has a direct link to D with cost 8.

Since 4 is less than 8, router A updates its routing table and chooses the path through B.

Step 3: Final routing information at router A

- Destination A: Cost = 0
- Destination B: Cost = 1, next hop = B
- Destination C: Cost = 3, next hop = B
- Destination D: Cost = 4, next hop = B

Therefore, the shortest path from A to D is:

$$\boxed{A \rightarrow B \rightarrow C \rightarrow D}$$

Total cost = 4.

This example shows that a path through intermediate routers can be cheaper than a direct link.

Advantages

1. Simple to understand and implement.
2. Requires less computational power.
3. Routers need to exchange information only with their neighbours.

Disadvantages

1. Convergence may be slow when the network topology changes.
2. It may create routing loops.
3. It suffers from the count-to-infinity problem, where routers may keep increasing the distance to an unreachable destination.

Example of a distance-vector routing protocol: RIP (Routing Information Protocol).

Question 11. Explain the Link-State Routing Algorithm and Describe Its Major Steps

Definition

The Link-State Routing Algorithm is a dynamic routing algorithm in which every router builds a complete map of the network topology and uses it to determine the shortest path to every destination.

It is also called the Shortest Path First (SPF) algorithm and uses Dijkstra's algorithm to calculate the shortest paths.

In link-state routing, routers share information about the state and cost of their directly connected links with all other routers in the network.

Major Steps of the Link-State Routing Algorithm

1. Neighbour Discovery

Each router identifies its directly connected neighbours using Hello packets. It also determines whether the links to these neighbours are active.

2. Measuring Link Costs

Each router determines the cost of reaching its directly connected neighbours. The cost may depend on factors such as bandwidth or delay.

3. Creating Link-State Packets (LSPs)

Each router creates a Link-State Packet containing:

- Router identification
- List of directly connected neighbours
- Cost of each link
- Sequence number and other control information

4. Flooding of Link-State Packets

Each router sends its LSP to its neighbours. The neighbours store the information and forward it to their neighbours.

This process, called flooding, continues until all routers in the network receive the link-state information.

5. Building the Network Topology Database

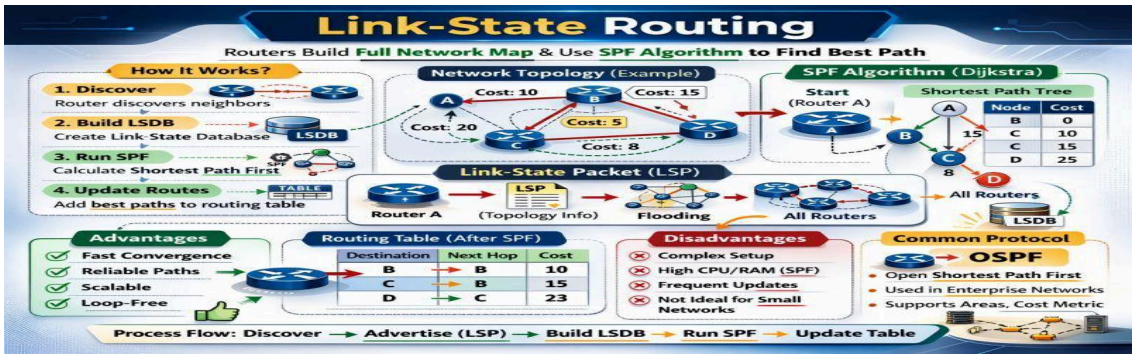
After receiving LSPs from all routers, each router constructs a complete map of the network. This map is stored in the Link-State Database (LSDB).

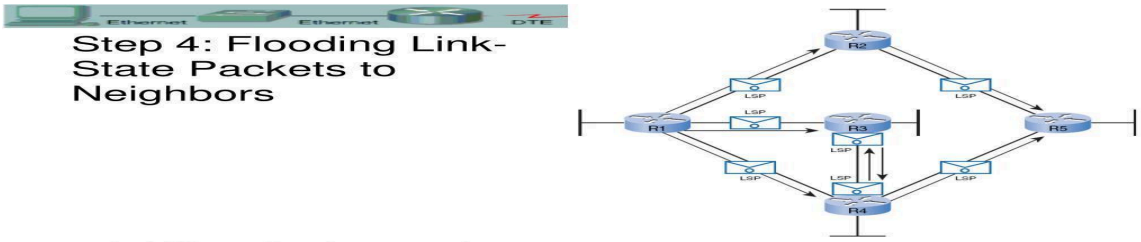
6. Shortest Path Calculation

Each router independently applies Dijkstra's algorithm to the network topology database to calculate the shortest path to every destination.

7. Updating the Routing Table

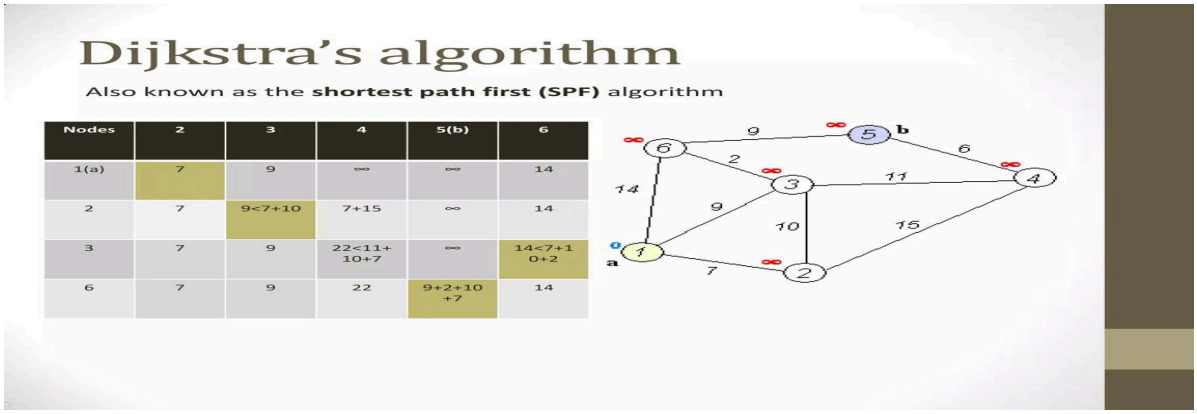
The router uses the calculated shortest paths to update its routing table, including the destination network, next hop, and total cost.





- An LSP needs to be sent only:
  - During initial startup of the router or of the routing protocol process on that router
  - Whenever there is a change in the topology,
    - link going down
    - link coming up
    - neighbor adjacency being established
    - neighbor adjacency being broken

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Example

Consider a network with routers A, B, C, and D.

Link costs:

- A–B = 2
- A–C = 5
- B–C = 1
- B–D = 4
- C–D = 2

To find the shortest path from A to D:

- Path A → B → D = 2 + 4 = 6
- Path A → C → D = 5 + 2 = 7
- Path A → B → C → D = 2 + 1 + 2 = 5

Therefore, the shortest path is:

A → B → C → D

Total cost = 5

Each router performs this calculation independently using the same network topology database.

Advantages

1. Provides fast convergence when the network topology changes.
2. Each router has a complete map of the network.
3. Reduces the possibility of routing loops.

4. Supports efficient shortest-path calculation.

Disadvantages

1. Requires more memory to store the topology database.
2. Requires more processing power to run Dijkstra's algorithm.
3. Flooding of link-state packets may consume bandwidth.

Example of a link-state routing protocol: OSPF (Open Shortest Path First).

Question 12. Explain the Structure of the Internet, Including ISPs and Their Interconnections

Structure of the Internet

The Internet is a global network of interconnected computer networks. It has evolved from a tree-like structure with a single backbone to a multi-backbone structure operated by different private companies.

The Internet consists of the following networks:

1. Backbone Networks: High-speed networks operated by private communication companies that provide global connectivity.
2. Peering Points: Points where different backbone networks connect and exchange traffic.
3. Provider Networks: Networks that use backbones for global connectivity and provide Internet services to customers.
4. Customer Networks: Networks that use the services provided by ISPs to access the Internet.

Internet Service Providers (ISPs)

An Internet Service Provider (ISP) is an organization that provides Internet connectivity to customers.

ISPs are interconnected to provide global communication. Their networks exchange data through backbone links and peering points.

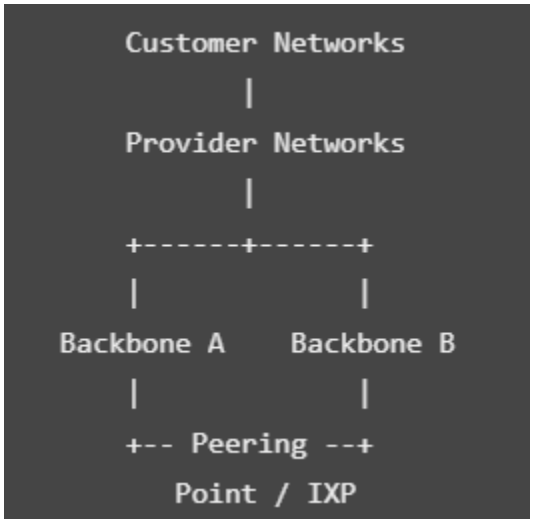
Autonomous System (AS)

Each ISP is considered an Autonomous System (AS) because it manages its own networks and routers.

- Intra-AS Routing: Routing within an AS using an Interior Gateway Protocol (IGP).
- Inter-AS Routing: Routing between different ASs using an Exterior Gateway Protocol (EGP).

This hierarchical routing structure allows each ISP to manage its own network while maintaining connectivity with the global Internet.

Diagram



Question 13. Design the Subblocks Using the Given Address Block

Given:

- Starting address: `14.24.74.0/24`
- Subnet 1 requires 120 addresses.
- Subnet 2 requires 60 addresses.
- Subnet 3 requires 10 addresses.

The given block contains:

$$2^{32-24} = 256 addresses$$

We allocate addresses from the largest subblock to the smallest subblock to avoid wastage and ensure proper alignment.

Step 1: Subblock for 120 Addresses

Since 120 is not a power of 2, allocate 128 addresses.

$$N_1 = 128 = 2^7$$

Prefix length:

$$n_1 = 32 - \log_2 128 = 25$$

- First address: `14.24.74.0/25`
- Last address: `14.24.74.127/25`

Step 2: Subblock for 60 Addresses

Allocate 64 addresses because 60 is not a power of 2.

$$N_2 = 64 = 2^6$$

Prefix length:

$$n_2 = 32 - \log_2 64 = 26$$

- First address: `14.24.74.128/26`
- Last address: `14.24.74.191/26`

Step 3: Subblock for 10 Addresses

Allocate 16 addresses because 10 is not a power of 2.

$$N_3 = 16 = 2^4$$

Prefix length:

$$n_3 = 32 - \log_2 16 = 28$$

- First address: `14.24.74.192/28`

- Last address: `14.24.74.207/28`

Final Answer

Required addresses	Allocated addresses	Prefix	Address range
120	128	`/25`	`14.24.74.0 – 14.24.74.127`
60	64	`/26`	`14.24.74.128 – 14.24.74.191`
10	16	`/28`	`14.24.74.192 – 14.24.74.207`

Unused addresses: 256 − (128 + 64 + 16) = 48 addresses.

The remaining addresses are from `14.24.74.208` to `14.24.74.255`.

Based on the worked example in your CNS notes, Module 2, page 19.

Question 14. IPv4 Addressing and CIDR

Given:

IP address: `192.168.10.0`

Number of computers: 200

a) Identify the Class of the Given IP Address

The first octet of the given IP address is 192.

Class C addresses range from 192 to 223.

Therefore, the given IP address belongs to Class C.

b) Determine the Default Subnet Mask

The default subnet mask for a Class C address is:

`255.255.255.0`

CIDR notation: `/24`

c) Is the Assigned Network Sufficient?

For a Class C network:

Total number of addresses:

$$2^8 = 256$$

Usable host addresses:

$$256 - 2 = 254$$

Two addresses are reserved:

- Network address
- Broadcast address

Since 254 usable host addresses are available and the college requires 200 computers, the assigned network is sufficient.

d) Suggest a Suitable CIDR Prefix and Determine the Number of Usable Host Addresses

The number of host bits required is calculated using:

$$2^h - 2 \geq 200$$

For 8 host bits:

$$2^8 - 2 = 254$$

Therefore:

- Host bits = 8
- Network bits = 32 − 8 = 24
- Suitable CIDR prefix = `/24`
- Subnet mask = `255.255.255.0`

Number of usable host addresses:

$$2^8 - 2 = 254$$

Final Answer: The suitable network is `192.168.10.0/24`, which provides 254 usable host addresses and is sufficient for 200 computers.

Question 15. Apply Dijkstra's Algorithm to Find the Shortest Path from A to E

Given link costs:

A–B = 2, A–C = 5, B–C = 1, B–D = 4, C–D = 2, C–E = 6, D–E = 1.

a) Identify the Shortest Path from A to E

Using Dijkstra's algorithm, the shortest path is:

A → B → C → D → E

b) Calculate the Total Cost

The total cost is:

$$2 + 1 + 2 + 1 = 6$$

Therefore, the shortest path cost = 6.

c) Explain How Dijkstra's Algorithm Determines the Shortest Path

Dijkstra's algorithm finds the shortest path from a source router to all other routers in a network.

1. Initialization: Set the distance of source router A to 0 and all other routers to infinity (∞).
2. Select Minimum Distance: Select the unvisited router with the smallest tentative distance.
3. Update Neighbours: Calculate the distance to each neighbouring router using:

New distance = Current distance + Link cost

4. Update Routing Information: If the calculated distance is smaller than the existing distance, update the distance and predecessor.
5. Repeat: Mark the selected router as visited and repeat the process until all routers are visited.
6. Shortest Path: Trace the predecessor information backwards from the destination E to obtain the shortest path.

d) If the Cost of Link D–E Increases to 10, Determine the New Shortest Path and Its Cost

After increasing the link cost D–E from 1 to 10, calculate the possible paths:

- A → B → C → D → E = 2 + 1 + 2 + 10 = 15
- A → B → C → E = 2 + 1 + 6 = 9
- A → B → D → E = 2 + 4 + 10 = 16
- A → C → E = 5 + 6 = 11

The minimum cost is 9.

Therefore, the new shortest path is A → B → C → E.

Total cost = 9.